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Clinical Application Project Report

Exploring Deep Learning Methodologies for Depression Detection

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Abstract

To address the subjectivity of traditional Major Depressive Disorder (MDD) diagnostics, this project investigates deep learning for depression detection using solely audio from clinical interviews.

This work compares two approaches on the DAIC-WOZ dataset: a baseline segment-level Convolutional Neural Network (CNN) with majority voting, and a hybrid patient-level CNN-Long Short-Term Memory (LSTM) model designed to capture temporal dynamics.

The results show the hybrid CNN-LSTM model significantly outperformed the baseline, achieving an F1-Score of 66.7% and an Unweighted Average Recall (UAR) of 73.24%, compared to the segment-level model's 51.35% and 59.70%, respectively.

This finding underscores that modeling the temporal evolution of vocal patterns is critical for this task, reinforcing the potential of automated speech analysis as a supplementary tool for objective mental health assessment.

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1 Introduction

Major Depressive Disorder (MDD) is a widespread and debilitating mental health condition, affecting millions globally [1]. It involves sadness, a loss of interest in previously enjoyed activities, and a range of other emotional, cognitive, and physical symptoms [2, 3]. The conventional diagnostic process uses clinical interviews and self-report questionnaires, with PHQ-9 being the most common one [4]. Although these methods are well-established, they are inherently subjective and depend on patients' ability and willingness to accurately report their symptoms. This subjectivity can lead to challenges in early detection and objective monitoring of treatment progress.

In the search for more objective biomarkers, researchers have increasingly turned to behavioral cues that can be captured and analyzed computationally [5]. Among these, speech has emerged as a promising, non-invasive indicator of an individual's psychological state [5, 6]. It has long been observed clinically that depression changes vocal patterns significantly [7]. Individuals with depression often experience psychomotor retardation, which directly impacts their speech. This can be exhibited as a slower tempo, more frequent and extended pauses, and a limited pitch variation that is perceived as a monotonous voice [3]. As these vocal characteristics can be measured, they can function as acoustic clues to an underlying depressive state. The automated analysis of these features offers a pathway to supplement traditional diagnostic methods with objective, scalable, and cost-effective screening tools [5].

Recent advancements in deep learning have provided robust new methodologies for analyzing complex data like audio. Unlike traditional machine learning, which relies on handcrafted features, deep learning models can automatically learn relevant hierarchical patterns directly from raw or minimally processed data [8]. This project explores the application of deep learning to detect depression solely from audio recordings of clinical interviews. It deliberately focuses on the acoustic dimension, motivated by two key factors. Firstly, audio data is less computationally intensive and raises fewer privacy concerns than video recordings. Secondly, isolating the audio modality allows for a focused investigation into the predictive power of vocal biomarkers alone, providing a clear baseline for understanding their contribution to depression detection.

To this end, this report investigates and compares two deep learning architectures. The first is a straightforward approach using a Convolutional Neural Network (CNN) to classify short audio segments, with patient-level decisions made via majority voting. It serves as a baseline to assess the effectiveness of localized acoustic patterns. The second, more sophisticated model is a hybrid CNN-Long Short-Term Memory (LSTM) network. This architecture represents a robust yet manageable approach. It enhances the powerful feature extraction of CNNs by integrating the sequential learning capabilities of LSTMs, allowing the model to effectively process patterns over time. The underlying logic is that depression

manifests not in fleeting acoustic events, but in sustained patterns across a conversation, such as persistent vocal flatness or variations in speech energy. Accordingly, the CNN-LSTM model is employed to specifically account for these crucial temporal features. By evaluating these two approaches on the publicly available DAIC-WOZ dataset [9], this project seeks to demonstrate the feasibility of audio-based depression detection and highlight the critical importance of modeling temporal context for achieving robust performance.

2 Background

2.1 Medical Background of Depression

Major Depressive Disorder (MDD) is a significant mood disorder characterized by a persistent feeling of sadness or a loss of interest in previously enjoyed activities [2]. According to the criteria established in the Diagnostic and Statistical Manual of Mental Disorders, a diagnosis of MDD requires the showing of at least five core symptoms over two weeks, representing a change from previous functioning. One of these symptoms must be either a depressed mood or a loss of interest [2]. Other symptoms include significant weight changes, decrease in sleep quality, fatigue, feelings of worthlessness, reduced ability to concentrate, and recurrent thoughts of death [2].

A key sign of MDD is a noticeable change in physical movement. This can manifest as either psychomotor agitation or psychomotor retardation, which involves a slowing down of thought, speech, and physical movement [2, 3]. These symptoms are subjective and observable to others and can significantly impact an individual's daily functioning. Such motor disturbances are considered a core feature of depression and can serve as essential diagnostic and therapeutic indicators [3].

The current standard for diagnosing depression typically involves a comprehensive psychiatric evaluation, which includes a structured clinical interview conducted by a mental health professional [10]. Together with these interviews, standardized questionnaires are widely used to assess the severity of depression. One of the most common and validated tools is the Patient Health Questionnaire-9 (PHQ-9), where patients self-report on nine items corresponding to the diagnostic criteria for major depressive disorder [4].

Each item is scored on a scale ranging from 0 ("not at all") to 3 ("nearly every day"). For instance, a patient is asked to rate how often they have been bothered by "little interest or pleasure in doing things" over the last two weeks. The individual scores for each of the nine questions are then summed to produce a total score ranging from 0 to 27. This total score is used to classify the severity of the depression [4]. The questionnaire corresponds directly to the diagnostic criteria for MDD listed in the DSM-5 and effectively monitors symptom severity [4].

For the DAIC-WOZ dataset used in this project, the ground truth for depression classification is established using the Patient Health Questionnaire-8 (PHQ-8), a slightly modified version of the PHQ-9. A score of 10 or greater on the PHQ-8 questionnaire is used to identify individuals with depression [10].

2.2 Speech as an Indicator of Depression

Speech is a sensitive biometric marker that reflects an individual's psychological condition. It's a long-standing clinical observation that individuals with depression frequently show changes in their vocal patterns, such as a slower rate of speech, reduced pitch variation, and articulation errors [6]. The automated speech analysis has thus emerged as a promising, non-invasive method for objectively quantifying these vocal biomarkers to aid in detecting and monitoring MDD [8].

Speaking requires the coordination of numerous muscle groups, a process influenced by changes in the nervous system [7]. The neurophysiological changes that come with depression can impair the neural pathways that govern speech. Alterations in the nervous systems that control muscle function can lead to abnormal tension, which directly interferes with the mechanics of phonation and articulation [7]. Consequently, this loss of precise motor control can audibly change the acoustic qualities and prosody of an individual's voice.

These changes manifest as specific, quantifiable characteristics in the speech signal. A primary indicator is a flatter pitch contour, perceived as monotonous speech. A significant differentiator is the pitch linear regression slope, with depressed individuals showing a flatter slope (less pitch variation over time) compared to healthy individuals [8]. While many studies report reductions in vocal intensity (loudness) [6], this can be context-dependent; the same study found that their cohort of depressed participants actually spoke significantly louder than healthy controls, a finding also noted in some earlier research [8, 11]. Furthermore, measures of voice quality reflecting vocal fold instability, such as jitter (irregularity in pitch) and shimmer (irregularity in amplitude), show consistent, moderate correlations with depression severity scores [8].

These vocal characteristics are the quantifiable manifestations of core depressive symptoms. The slowing of speech, increased pausing, and longer utterance durations are directly linked to psychomotor retardation (PMR), a known indicator of MDD [12, 13]. The reduced pitch range and flatter tone of voice also serve as a vocal marker for the emotional numbness of depression. This vocal pattern audibly reflects the blunted affect and anhedonia (the loss of pleasure and emotional expression) common in MDD [14].

Subsequently, analyzing these distinct vocal patterns makes it possible to capture objective, data-driven indicators of an individual's underlying depressive state, with machine learning models based purely on acoustic features [8].

2.3 Deep Learning for Audio Analysis

A common and highly effective approach is to transform the raw audio into a two-dimensional representation. This strategy allows researchers to leverage the powerful and well-established 2D CNN architectures that have excelled at image recognition tasks. For this project, we adopt this latter approach by converting the audio into a mel spectrogram, a visual representation of how the spectral content of a signal evolves. [15]. The process involves applying a Short-Time Fourier Transform (STFT) on very short time segments and analyze the frequency spectrum

of each segment. The resulting frequencies are subsequently mapped to the Mel-scale. This is a perceptual scale that organizes pitches in a way that corresponds to how people judge their relative distance [16]. This transformation is particularly effective because the Mel-scale approximates the non-linear frequency response of the human ear, making the resulting spectrogram more perceptually relevant to how humans hear audio [16, 15].

2.3.1 Convolutional Neural Networks for Spectrogram Analysis

Convolutional Neural Networks (CNNs) are a class of deep neural networks that are highly effective at analyzing visual data. By representing audio visually in the form of Mel-spectrograms, we can leverage the powerful pattern-recognition capabilities of CNNs, which have been widely adapted for audio classification tasks [17, 18, 19]. In this context, a CNN treats the spectrogram as a 2D image, where the horizontal and vertical axes correspond to time and Mel-frequency respectively, and pixel intensity represents the signal's energy. The network applies a series of convolutional filters to the spectrogram to learn hierarchical features. Early layers in the network might learn to detect simple, low-level features such as basic textures or tonal shifts. In contrast, deeper layers can combine these to recognize more complex and abstract acoustic patterns [17, 18]. This hierarchical learning process allows CNNs to effectively model the underlying structures within audio data that may indicate depressive speech patterns.

2.3.2 Long Short-Term Memory Networks for Temporal Feature Integration

While CNNs excel at identifying local features within audio segments, they do not inherently model the temporal dependencies that exist across more extended sequences. To capture these longer-term patterns, this project employs Long Short-Term Memory (LSTM) networks, a specialized type of Recurrent Neural Network (RNN). LSTMs were explicitly designed to address the challenges of learning long-range dependencies in sequential data, a common issue in standard RNNs caused by the vanishing gradient problem [20].

They achieve this through a gating mechanism, comprising input, output, and forget gates, that regulates the flow of information through the network over time. This allows the LSTM to selectively remember or forget information from previous time steps, enabling it to model the temporal dynamics of a sequence effectively [20].

In the context of speech recognition and analysis, LSTMs have demonstrated significant success in capturing the contextual information exhibited in spoken language over time [21]. By processing the sequence of feature vectors extracted by the CNN, the LSTM can learn how vocal characteristics evolve over the course of an entire interview, providing a more holistic basis for a patient-level depression classification.

3 Methodology

3.1 Dataset

The data utilized in this study is sourced from the Distress Analysis Interview Corpus (DAIC-WOZ), a component of the Extended DAIC corpus [9]. This dataset was specifically curated for research on the diagnosis of psychological distress conditions, including depression, anxiety, and post-traumatic stress disorder, through automated analysis of clinical interviews [9]. The interviews are conducted by a virtual agent named Ellie, who engages participants in a series of questions designed to bring out verbal and non-verbal indicators of mental health status. These interactions were managed by a human clinician in a Wizard-of-Oz (WOZ) setup, ensuring that the interviews remained naturalistic and clinically relevant [9].

The DAIC-WOZ corpus is multimodal, containing synchronized audio recordings, video footage, and transcripts of the interviews. However, only English audio data was employed for analysis in this project. The dataset contains interviews from 189 unique participants, partitioned into a training set of 107 individuals, a development set of 35, and a test set of 47 [9].

The ground truth for the binary depression classification task is established using the Patient Health Questionnaire (PHQ-8), a standard self-report measure for assessing the severity of depressive symptoms. In line with standard clinical practice and the original dataset design, a participant is assigned to the 'depressed' class if their PHQ-8 score is 10 or greater, and to the 'non-depressed' class otherwise [9]. The training set comprises 30 participants with depression and 77 without, while the development set contains 12 depressed and 23 non-depressed individuals.

3.2 Data Preprocessing

Before model training, a series of preprocessing steps was applied to the raw audio data from the DAIC-WOZ corpus to convert it into a suitable format for the deep learning architectures. This involved segmenting the lengthy interview recordings into shorter chunks and then transforming these chunks into a two-dimensional spectrogram representation that captures the essential acoustic features of speech.

3.2.1 Audio Segmentation

The complete audio recordings of the clinical interviews, which vary in length, are too long to be processed by a neural network. Therefore, the audios were divided into smaller, uniform

segments. Each patient’s complete interview audio was partitioned into 5-second segments with 1-second overlap for this project.

This segment length was chosen to balance two factors: The segments must be long enough to capture meaningful paralinguistic and prosodic features, such as intonation, speech rate, and pauses, that may indicate depressive states. At the same time, they must be short enough to be computationally manageable and to provide a sufficient number of samples for training the segment-level models [22].

3.2.2 Spectrogram Generation Parameters

To generate the Mel-spectrograms for each 5-second audio segment, the following parameters were used:

- **Sample Rate:** The audio was processed at a sample rate of 16,000 Hz.
- **Window Size:** A window size of 512 samples was used for the STFT, defining the length of each frame for analysis.
- **Hop Size:** A hop size of 160 samples determined the step size, or overlap, between consecutive frames.
- **Mel-Bins:** The frequency spectrum was mapped onto 64 Mel-bins.
- **Frequency Range:** The analysis focused on the frequency range from 50 Hz (f_{min}) to 8000 Hz (f_{max}), which covers the primary range of human speech.

The resulting spectrograms were then converted to a logarithmic decibel (dB) scale. This process yields a 2D tensor for each audio segment, which serves as the direct input to the CNN model.

3.3 Model Architectures

To investigate the detection of depression from audio recordings, this project compares two distinct deep learning methodologies. The first approach operates at the segment level, classifying individual chunks of audio and aggregating the results. The second operates at the patient level, leveraging a hybrid architecture to model temporal dependencies across an entire interview. Both models use the CNN10 architecture as a foundational component [19].

3.3.1 Segment-Level CNN with Majority Voting

As seen in figure 3.1, the first model employs a CNN to classify individual audio segments. By representing audio segments as Mel-spectrograms, we can treat the audio classification task as an image recognition problem, where the network learns to identify patterns in the time-frequency representation indicative of depression.

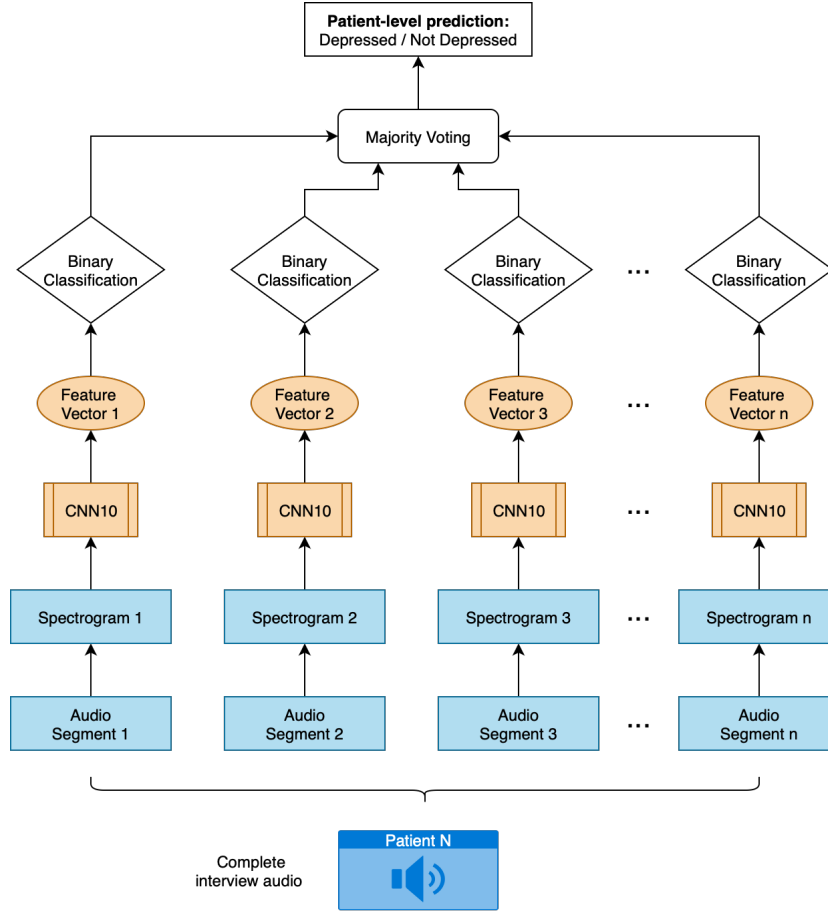


Figure 3.1: Segment-Level CNN Network with Majority Voting

The specific architecture used is the CNN10, a 10-layer network that has demonstrated strong performance on various audio classification tasks [19]. This model processes each Mel-spectrogram independently, passing it through convolutional layers that extract hierarchical features. These layers learn to detect low-level features like textures and edges in the spectrogram (corresponding to acoustic events) in the initial layers, and more complex, abstract features in deeper layers. The output for each segment is a binary prediction: 'depressed' or 'not depressed'.

Since the ground truth labels provided in the DAIC-WOZ dataset are at the patient level, a mechanism is required to aggregate the predictions from all segments belonging to a single patient. For this model, a hard voting, or majority voting, scheme is implemented. This straightforward and effective method counts all the binary predictions for a patient's segments, and the class that receives the most votes becomes the final patient-level classification.

3.3.2 Patient-Level CNN-LSTM

The second methodology adopts a more holistic, patient-level approach by combining CNNs with an LSTM network. This hybrid CNN-LSTM architecture is designed to capture the acoustic characteristics within short audio segments and the temporal dynamics that unfold over an entire clinical interview [23]. We incorporate LSTM layers to capture the temporal dynamics of the input data, since LSTMs are designed to model sequential dependencies by maintaining a memory of past information through gating mechanisms [20]. This makes them particularly effective in tasks such as depression detection, where the temporal evolution of speech and behavioral cues is critical. Such temporal patterns, like changes in speech rate or prosody, may contain crucial diagnostic information that a segment-level model might miss.

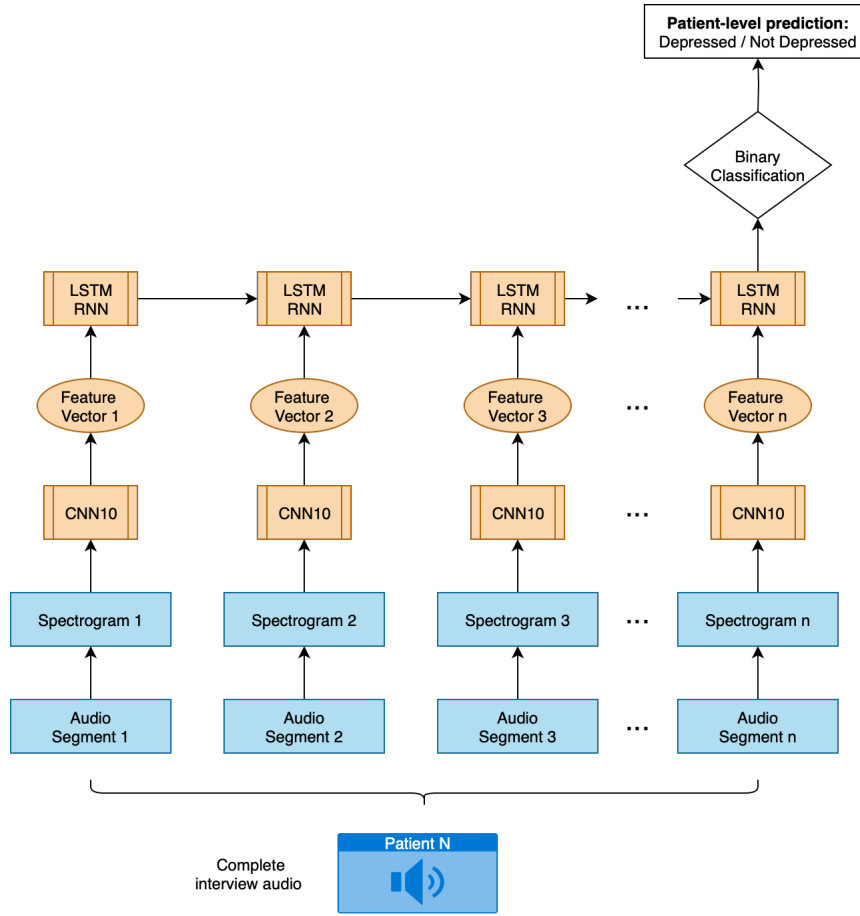


Figure 3.2: Patient-Level CNN-LSTM Network

As can be seen in figure 3.2, the process involves two main stages:

1. **Feature Extraction:** The CNN10 model, pre-trained on the audio classification task, is repurposed as a feature extractor. For each patient's audio segment (Mel-spectrogram), the CNN processes the input and, instead of producing a final classification, outputs

a dense feature vector from its second-to-last layer. This vector serves as a compact, learned representation of the acoustic information within that specific segment.

2. **Sequence Classification:** The sequence of feature vectors generated from all patient interview segments is then fed into an LSTM layer. The LSTM processes the entire sequence of feature vectors, learning to recognize patterns and shifts in vocal characteristics over time. The final hidden state of the LSTM, which summarizes the information from the whole sequence, is then passed to a classifier head, a dense layer with a sigmoid activation function, to make a single, patient-level prediction of depression. This approach allows the model to base its final decision on an understanding of the entire interview’s temporal and acoustic properties.

3.4 Experimental Setup & Evaluation

The open-source autrainer toolkit was utilized to structure and manage the entire experimental pipeline, from data preprocessing to model training and evaluation [24]. Autrainer is built on PyTorch and Hydra, provides a command-line interface (CLI) and a programmatic wrapper to handle configurations, data processing, training, and post-processing, ensuring a reproducible and structured workflow [24]. For training the neural networks, the Binary Cross-Entropy (BCE) loss function was employed. BCE is a standard loss function for binary classification tasks, as it effectively measures the dissimilarity between the model’s predicted probabilities and the true binary labels (0 or 1) [25].

Model parameters were optimized using the Adam optimizer [26], an adaptive learning rate optimization algorithm that is suited for a wide range of deep learning tasks due to its computational efficiency and low memory requirements. The initial learning rate was set to 0.001, a commonly used default value that provides a good starting point for convergence.

The evaluation of classification models in clinical domains requires metrics that are robust to class imbalance and reflect real-world diagnostic priorities [27]. The DAIC-WOZ dataset exhibits a significant imbalance between the ‘depressed’ (minority class) and ‘non-depressed’ (majority class) participants. In such scenarios, Accuracy can be a highly misleading metric, as a naive model that consistently predicts the majority class would achieve a high score while being clinically useless [28].

To provide a clinically meaningful assessment, this study employs a suite of metrics focused on performance for each class independently. The primary emphasis is placed on Unweighted Average Recall (UAR) and the F1-Score.

- **Recall (Sensitivity):** This measures the model’s ability to correctly identify true positive cases ($Recall = \frac{TP}{TP+FN}$). In a depression screening context, high recall is paramount. A false negative (failing to identify a depressed individual) has far more severe potential consequences than a false positive (incorrectly flagging a healthy individual for further screening) [28]. Therefore, maximizing recall for the ‘depressed’ class is a critical objective.

- **Precision:** This measures the proportion of positive predictions that were actually correct ($Precision = \frac{TP}{TP+FP}$). While recall is crucial, precision is also important for assessing a model's reliability and minimizing the burden of unnecessary follow-up assessments on both patients and clinicians [28].
- **F1-Score:** The F1-Score is the harmonic mean of Precision and Recall ($F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$). It provides a single, consolidated metric that balances the trade-off between finding as many depressed individuals as possible (Recall) and ensuring that the individuals identified are indeed depressed (Precision). It is particularly valuable for imbalanced datasets where both false positives and false negatives carry costs [25].
- **Unweighted Average Recall (UAR):** UAR is the arithmetic mean of the recall scores for each class. In this binary case, it is the average of the recall for the 'depressed' class (sensitivity) and the recall for the 'non-depressed' class (specificity). UAR gives equal weight to the performance on the minority and majority classes, making it one of the most reliable metrics for evaluating overall performance on imbalanced datasets [29]. A high UAR indicates that the model is effective at identifying both depressed and healthy individuals, which is the goal of a robust screening tool.

This evaluation strategy ensures a comprehensive understanding of the models clinical utility, going further than simple accuracy to assess their effectiveness in a realistic screening scenario.

4 Experiments and Results

4.1 Experiments & Results

This section details the outcomes of the experimental evaluation. It compares performance of the two proposed models: the segment-level CNN with majority voting and the patient-level CNN-LSTM. All models were evaluated on the held-out test set of the DAIC-WOZ corpus [9, 30].

4.1.1 Segment-Level CNN with Majority Voting

The initial approach, which classifies aggregated audio segments via majority vote, serves as our baseline. Table 4.1 displays the performance metrics for this model. The model achieved an F1-Score of 51.35% and an Unweighted Average Recall (UAR) of 59.70%. While the model correctly identified a high percentage of depressed individuals (Recall of 84.6%), this came at the cost of misclassifying many healthy individuals, as indicated by the lower Precision (52.2%) and Specificity (34.8%). This suggests a tendency for the model to predict the ‘depressed’ class.

Table 4.1: Performance of the Segment-Level CNN with Majority Voting Model

Metric	Value (%)
Accuracy	55.22
Precision	52.20
Sensitivity	84.62
Specificity	34.78
F1-Score	51.35
UAR	59.70

Table 4.2: Performance of the Patient-Level CNN-LSTM Model

Metric	Value (%)
Accuracy	72.2
Precision	58.82
Sensitivity	76.92
Specificity	69.57
F1-Score	66.7
UAR	73.24

4.1.2 Patient-Level CNN-LSTM Model

To capture the temporal dynamics within an entire interview, the second model utilized a CNN as a feature extractor, the outputs of which were then fed into an LSTM for sequence-level classification. Table 4.2 presents the results for this integrated CNN-LSTM architecture. This model demonstrated a notable improvement over the segment-level approach, achieving an F1-Score of 66.7% and a UAR of 73.24%. Notably, this model provides a much better balance between identifying depressed (Recall of 76.9%) and non-depressed (Specificity of

69.6%) participants. The significant increase in Precision to 58.8% indicates a reduction in false positive predictions compared to the baseline.

The results indicate that the patient-level CNN-LSTM model, which considers the sequential nature of the interview data, outperforms the segment-level model that relies on a simple majority vote. The improved F1-Score and UAR suggest that modeling temporal patterns is crucial for effectively detecting depression from audio recordings. Both models performed considerably better than random chance, indicating that they have learned significant predictive patterns from the audio data.

5 Discussion

5.1 Interpretation of Results

The experimental results demonstrate the superior performance of the patient-level CNN-LSTM model over the segment-level CNN with majority voting. The final model yielded a substantial improvement in both F1-Score (66.7% vs. 51.35%) and Unweighted Average Recall (UAR) (73.24% vs. 59.70%). This outcome strongly suggests that incorporating temporal context is critical for effectively identifying depression from audio interviews.

The primary limitation of the segment-level model lies in its approach to aggregation. The model discards the sequential nature of speech by classifying each audio segment in isolation and relying on a simple majority vote. This method cannot capture the long-term, dynamic patterns in vocal characteristics that often signify depression, such as sustained monotone pitch, consistently low energy, or the rhythmic patterns of speech and pauses over the course of a conversation [31]. The model effectively treats the interview as an unordered collection of audio clips and, therefore, loses valuable contextual information revealed over time.

In contrast, the CNN-LSTM architecture was specifically designed to overcome this limitation. The CNN component acts as a feature extractor, identifying local patterns within each Mel-spectrogram. The LSTM layer processes the sequence of these feature vectors, enabling the model to learn and model the long-range temporal dependencies between them [32]. This ability to analyze how vocal features evolve and interrelate throughout an entire interview is likely the reason for its enhanced performance. The architecture’s success highlights the importance of modeling sequential information in speech for depression and emotion recognition [32].

From a clinical perspective, the metric scores offer further insight. The F1-Score provides a balanced measure of a classifier’s performance by considering false positives and false negatives. The notable increase in the CNN-LSTM’s F1-Score suggests it achieves a better trade-off between correctly identifying depressed individuals (recall) and not misclassifying healthy individuals (precision). This is crucial in screening to minimize missed cases and unnecessary follow-ups. Furthermore, the higher UAR confirms that the model is more effective at identifying both depressed and non-depressed participants, reducing the risk of bias toward the majority class. While the performance is considerably better than random chance and demonstrates that the model has learned significant predictive patterns, the scores also indicate that the model is not yet robust enough for standalone diagnostic use and should only be considered a supplementary tool for clinical assessment.

5.2 Comparison with Existing Work

The performance of our patient-level CNN-LSTM model, which achieved an F1-Score of 66.7%, can be contextualized by comparing it to several other deep learning approaches applied to the same dataset as seen in Table 5.1.

Table 5.1: Comparison of F1-Scores for Depression Detection on the DAIC-WOZ Dataset

Work	Approach	F1-Score (%)
Zhang et al. (2024)	wav2vec 2.0 + LSTM	79.0
Yue et al. (2024)	DWAM-Former	78.8
This Work	Patient-Level CNN-LSTM	66.7
Vázquez-Romero et. al. (2020)	1D-CNN Ensemble	65.0
This Work	Segment-Level CNN + Voting	51.35

A relevant baseline is the work by Vázquez-Romero and Gallardo-Antolín (2020), who proposed an ensemble of 50 individual 1D-CNN models. Their approach processes log-spectrograms but, similar to our initial segment-level model, does not explicitly model long-range temporal sequences, instead relying on the fusion of multiple independent classifiers. Their best-performing ensemble achieved an F1-Score of 65.0% for the depressed class [33].

More recent and complex approaches have demonstrated this task’s upper bounds of performance. Zhang et al. (2024) leveraged transfer learning by using the powerful wav2vec 2.0 model [34], pre-trained on vast amounts of unlabeled speech, as a feature extractor. By feeding these rich features into a downstream model composed of a 1D-CNN, LSTM, and self-attention mechanisms, they achieved a state-of-the-art F1-Score of 79.0% [35]. Similarly, Yue et al. (2024) introduced DWAM-Former, a hierarchical transformer model that uses dynamic windowing to explicitly model speech structure from frames to words. Using features from another large pre-trained model, HuBERT, they reached a macro F1-score of 78.8% [36]. These results highlight that the most significant performance gains currently stem from advanced, pre-trained feature extractors and sophisticated attention-based architectures.

In summary, our CNN-LSTM model demonstrates a clear advantage over non-temporal ensemble methods and confirms the importance of sequential modeling. While it does not match the performance of state-of-the-art systems employing large-scale pre-training or highly complex transformer architectures, it establishes a solid performance baseline and validates our core architectural design.

5.3 Limitations of the Project

While this project successfully demonstrates the feasibility of using deep learning for depression detection from audio, it is important to acknowledge its limitations. These limitations include methodological choices, the dataset’s nature, and the findings’ generalizability.

5.3.1 Methodological Limitations

The CNN10 architecture is a strong baseline in audio classification. It is simpler than many state-of-the-art models already discussed that incorporate attention mechanisms, which have been shown to improve performance in similar tasks [32]. A more complex feature extractor could have yielded more discriminative features. For instance, the Audio Spectrogram Transformer (AST), proposed by Deng et al., offers a more advanced approach. This model is a convolution-free, purely attention-based architecture that processes audio spectrograms to capture long-range, global context [37].

Furthermore, the aggregation method in our initial model, majority voting, is a naive approach that discards all temporal information, a key limitation that motivated the development of the second model. The patient-level model, while superior, also has room for improvement. Segmenting the audio and feeding the resulting feature sequence into an LSTM is just one way to model temporal dynamics. Alternative approaches, such as using attention mechanisms to weigh the importance of different speech segments, might have captured more distinct patterns indicating depression [32, 37].

5.3.2 Dataset Limitations

Despite being a standard benchmark, the DAIC-WOZ dataset presents several limitations. Firstly, its size is relatively small for training deep learning models from scratch, which increases the risk of overfitting. Furthermore, the number of non-depressed cases is over four times that of depressed patients, which may favor non-depressed patients in classification [38]. Although our models learned significant patterns, their generalization ability may be constrained by the diversity of examples seen during training.

Another significant limitation is the dataset’s cultural and linguistic specificity. The interviews were conducted with an American population in English, and the vocal biomarkers of depression can vary across different cultures, languages, and dialects [39]. Consequently, the model has likely learned patterns specific to this demographic and may show bias or perform poorly when applied to individuals from different backgrounds. The findings from this project should not be considered universally applicable, and any real-world application would necessitate extensive testing and validation on diverse populations [39].

6 Conclusion

This project successfully demonstrated the viability of using deep learning methodologies for detecting depression from audio recordings of clinical interviews. By comparing a segment-level CNN with a patient-level hybrid CNN-LSTM model, our findings underscore the critical importance of incorporating temporal context in the analysis of speech for mental health assessment. The superior performance of the CNN-LSTM architecture, which achieved an F1-Score of 66.7% and a UAR of 73.24%, clearly indicates that dynamic vocal patterns unfolding over the course of an interview are more powerful predictors of depression than acoustic features from isolated speech segments.

The results confirm that models capable of learning long-range dependencies in speech can more effectively capture the sustained changes in pitch, energy, and rhythm characteristic of a depressive state. While the performance of our model is promising and significantly better than random chance, it does not yet reach the levels of state-of-the-art systems that employ large, pre-trained models or complex transformer architectures. Nevertheless, it serves as a robust validation of our architectural choices and establishes a solid baseline for future work.

From a clinical standpoint, this work contributes to the growing body of evidence supporting the use of automated speech analysis as a supplementary tool for mental health screening. Such technology could one day help clinicians make more objective assessments, monitor patient progress with greater frequency, and identify at-risk individuals earlier. However, the limitations of this study, particularly those related to the small and culturally specific dataset and the insufficient accuracy for a reliable classification, mean that the model is not yet ready for standalone diagnostic use.

Future research should focus on several promising avenues. Integrating other modalities of communication, such as the linguistic content from interview transcripts or the visual information from video recordings, could provide a more holistic and accurate picture of a patient's mental state [40, 41]. Additionally, exploring more advanced attention-based architectures and leveraging larger, more diverse datasets will be crucial for improving model robustness and ensuring its generalizability across different populations. Ultimately, this project reinforces the potential of voice as a digital biomarker and paves the way for more sophisticated and reliable AI-driven tools in mental healthcare.

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